Syed Shayan Mazahir

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Technical Skills

Languages: C/C++, Python, JavaScript, Java
Hardware & Embedded: Arduino, ESP32, Raspberry Pi, VEX, UART/Serial, I^2C, PWM, ADC, Ultrasonic/IR/Gyro/EMG Sensors, Servo & Motor Control, PID, Circuit Design, Oscilloscope & Multimeter
Tools & Libraries: Linux, OpenCV/MediaPipe, Digital Filtering & Signal Acquisition, 3D Printing (FDM), Git/GitHub, Node.js, Flask, REST APIs

Projects

ATOM — Humanoid Robot, Phase 0 Complete | *C++, Python, ESP32, MediaPipe, UART, ADC*
- Built a real-time gesture-mirroring robotic hand end to end: 21-landmark hand tracking → dot-product joint angles → per-user open/fist calibration → 5 servos over UART at 20 Hz
- Authored ESP32 firmware (C++) and a config-driven Python pipeline (shared JSON protocol) with rolling-average smoothing, rate limiting, and range clamping on both ends of the serial link
- Debugged a non-functional EMG acquisition chain from first principles: derived the expected ADC baseline from the 1.5 V reference, identified 20 Hz undersampling as aliasing, and isolated USB/mains coupling as the dominant noise source via single-variable testing (96% → 15% rail clipping)
- Found two defects in the vendor's published example code (16-bit `int` overflow on AVR; integer-division truncation of the signal envelope); maintained an 800-line experiment log with null controls and retracted results
- Designed a 5V/3A power architecture with a 2.5A servo budget and isolated logic/actuator supplies; published a web-based STL browser for the 3D-print files via GitHub Pages

Mr. Beep-A-Lot — Multi-Mode Autonomous Robot | *C++, Arduino, Processing (Java), I2C, Serial*
- Built an autonomous robot with three operational modes: obstacle avoidance, environment scanning, and human following
- Integrated IR remote control, dual ultrasonic sensors, a servo-controlled radar, and an I2C LCD display, with sensor fusion driving autonomous navigation
- Developed a Processing (Java) visualization for real-time radar data over Serial communication

games.random — AI-Powered Game Development Platform | *JavaScript, Claude API, Monaco Editor*
- Engineered few-shot prompt templates for the Claude API, achieving 85%+ syntactically correct code generation from natural language descriptions
- Built a real-time validation pipeline, reducing runtime errors by 40% via instant syntax highlighting and error feedback
- Open-sourced on GitHub with documentation of the LLM prompt engineering techniques used

Experience

Robotics Team Lead & Programming Hub Mentor — 2024 – 2025
Louise Arbour Secondary School — *Brampton, ON*
- Led a 15-student team to First Place in the RobOlympics competition
- Built 2 competition robots: a Bluetooth-controlled soccer robot with electronic braking (1st Place) and an autonomous line follower using proportional control with lost-line recovery (3rd)
- Integrated ultrasonic/IR sensors, L298N motor drivers, and HC-05 Bluetooth in C++ on Arduino; documented lessons learned on sensor calibration under competition lighting
- Engineered voltage regulation circuits (5V to 15V), electrical schematics, and wiring harnesses; specified and documented custom mechanical assemblies
- Built a Flask web application with Google API integration serving 500+ students, including OAuth 2.0 authentication and SSL/HTTPS
- Mentored 20+ students in Python and data structures, trained 15 in embedded C++ and circuit debugging with oscilloscopes and multimeters, and maintained annotated schematics and Git repositories

Education

University of Waterloo — Waterloo, ON
Bachelor of Geospatial Data Science, Minor in Computer Science — *Expected June 2030*